

Bayesian Learning

Lecture 11 - Bayesian model comparison

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Overview

- Bayesian model comparison
- Marginal likelihood
- Model averaging

Bayesian model comparison

■ Posterior model probabilities

$$\underbrace{\Pr(M_k|\mathbf{y})}_{\text{posterior model prob.}} \propto \underbrace{p(\mathbf{y}|M_k)}_{\text{marginal likelihood}} \cdot \underbrace{\Pr(M_k)}_{\text{prior model prob.}}$$

■ The **marginal likelihood** for model M_k with parameters θ_k

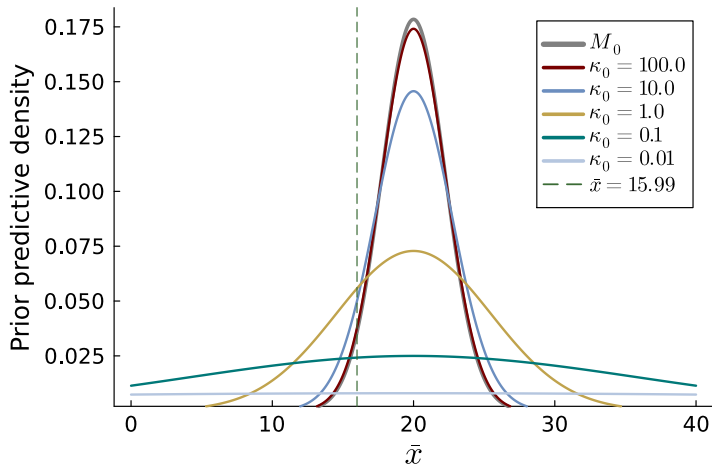
$$\underbrace{p(\mathbf{y}|M_k)} = \int p(\mathbf{y}|\theta_k, M_k)p(\theta_k|M_k)d\theta_k.$$

■ θ_k is 'removed' by the averaging wrt prior. **Priors matter!**

■ The **Bayes factor**

$$B_{12}(\mathbf{y}) = \frac{p(\mathbf{y}|M_1)}{p(\mathbf{y}|M_2)}$$

Internet speed data - prior predictive density



Laplace approximation

- Taylor approximation of the log posterior around the posterior mode $\tilde{\theta}$ gives:

$$\begin{aligned} p(\mathbf{y}|\theta)p(\theta) &\approx p(\mathbf{y}|\tilde{\theta})p(\tilde{\theta}) \exp \left[-\frac{1}{2} J_{\mathbf{y}}(\tilde{\theta})(\theta - \tilde{\theta})^2 \right] \\ &= p(\mathbf{y}|\tilde{\theta})p(\tilde{\theta})(2\pi)^{p/2} \left| J_{\mathbf{y}}^{-1}(\tilde{\theta}) \right|^{1/2} \\ &\quad \underbrace{\times (2\pi)^{-p/2} \left| J_{\mathbf{y}}^{-1}(\tilde{\theta}) \right|^{-1/2} \exp \left[-\frac{1}{2} J_{\mathbf{y}}(\tilde{\theta})(\theta - \tilde{\theta})^2 \right]}_{\text{multivariate normal density}} \end{aligned}$$

- So integrating both sides with respect to θ gives

$$p(\mathbf{y}) = \int p(\mathbf{y}|\theta)p(\theta)d\theta = p(\mathbf{y}|\tilde{\theta})p(\tilde{\theta})(2\pi)^{p/2} \left| J_{\mathbf{y}}^{-1}(\tilde{\theta}) \right|^{1/2}$$

Laplace approximation

- **The Laplace approximation** of the log marginal likelihood:

$$\ln \hat{p}(\mathbf{y}) \approx \ln p(\mathbf{y}|\tilde{\theta}) + \ln p(\tilde{\theta}) + \frac{1}{2} \ln \left| J_{\mathbf{y}}^{-1}(\tilde{\theta}) \right| + \frac{p}{2} \ln(2\pi),$$

where p is the number of unrestricted parameters.

- $\hat{\theta}$ and $J_{\mathbf{y}}(\tilde{\theta})$ can be obtained with **optimization/autodiff**.

Model averaging

- Let γ be a quantity with the same interpretation in the two models, M_1 and M_2 .
- Example: Prediction $\gamma = y_{T+1}$.
- The marginal posterior distribution of γ is

$$p(\gamma|\mathbf{y}) = p(M_1|\mathbf{y})p_1(\gamma|\mathbf{y}) + p(M_2|\mathbf{y})p_2(\gamma|\mathbf{y}),$$

$p_k(\gamma|\mathbf{y})$ is the marginal posterior of γ from model M_k .

- Predictive distribution includes **three sources of uncertainty**:
 - ▶ **Future errors**/disturbances (e.g. the ε 's in a regression)
 - ▶ **Parameter uncertainty** (the predictive distribution has the parameters integrated out by their posteriors)
 - ▶ **Model uncertainty** (by model averaging)